

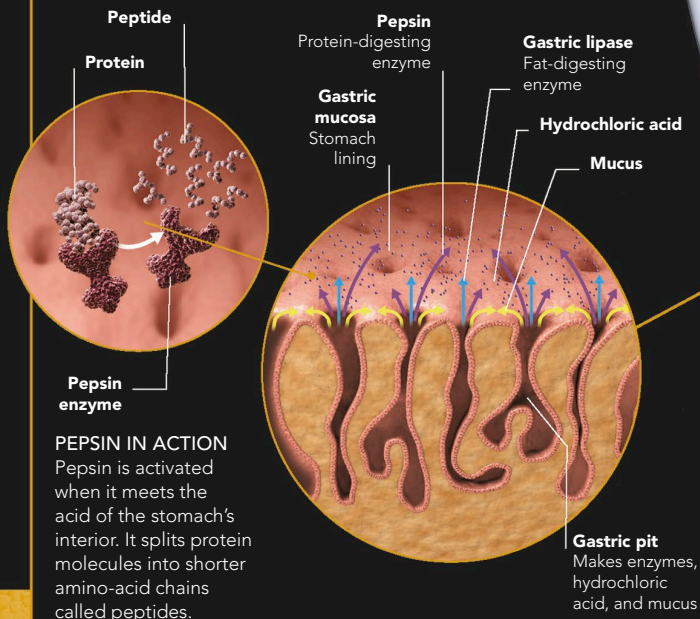
DIGESTION

THE DIGESTIVE PROCESS INVOLVES A SERIES OF PHYSICAL AND CHEMICAL ACTIONS THAT BREAK DOWN FOOD INTO NUTRIENT PARTICLES SMALL ENOUGH FOR ABSORPTION.

Vigorous physical digestion of food – mashing and churning – starts in the mouth and continues in the stomach using muscular movement. The stomach and the mouth secrete digestive chemicals (enzymes), too. By the time the pulverized food and enzymes (chyme) reach the duodenum, many food particles are already microscopically small, yet not small enough to pass across cell membranes into the body tissues. Chemical digestion in the small intestine splits large molecules into even smaller, absorbable particles that can enter the blood.

1 IN THE STOMACH

The stomach lining has gastric pits that make various substances: hydrochloric acid to kill microbes in swallowed food; the enzyme gastric lipase to begin breaking down fat; mucus to protect the stomach from digestive enzymes and acid; and pepsin to digest proteins. Pepsin on its own could digest the stomach wall, so it is first released in an inactive form (pepsinogen), then activated by the stomach's acid.



DIGESTIVE JOURNEY

As it travels from the mouth to the small intestine, food is broken down into smaller and smaller particles.